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$$f(x) = \ln(x^2 + \sqrt{x^4 - 1})$$

$$\begin{aligned}
f'(x) &= \frac{1}{x^2 + \sqrt{x^4 - 1}} \cdot \left(2x + \left(\frac{1}{2\sqrt{x^4 - 1}} \cdot 4x^3\right)\right) \\
&= \frac{1}{x^2 + \sqrt{x^4 - 1}} \cdot \left(2x + \frac{2x^3}{\sqrt{x^4 - 1}}\right) \\
&= \frac{1}{x^2 + \sqrt{x^4 - 1}} \cdot \left(\frac{2x\sqrt{x^4 - 1}}{\sqrt{x^4 - 1}} + \frac{2x^3}{\sqrt{x^4 - 1}}\right) \\
&= \frac{1}{x^2 + \sqrt{x^4 - 1}} \cdot \left(\frac{2x\sqrt{x^4 - 1} + 2x^3}{\sqrt{x^4 - 1}}\right) \\
&= \frac{1}{x^2 + \sqrt{x^4 - 1}} \cdot \left(\frac{2x(x^2 + \sqrt{x^4 - 1})}{\sqrt{x^4 - 1}}\right) \\
&= \frac{2x(x^2 + \sqrt{x^4 - 1})}{(x^2 + \sqrt{x^4 - 1})\sqrt{x^4 - 1}} \\
&= \frac{2x}{\sqrt{x^4 - 1}}
\end{aligned}$$

References